

5 Insulin is a peptide hormone produced by the pancreas.

Insulin plays a crucial role in regulating blood glucose levels and maintaining homeostasis in the body.

(a) State what is meant by the term **homeostasis**.

(1)

The effect of exercise on the concentration of insulin in the blood of trained and untrained individuals was investigated.

The graph compares the changes in insulin levels in an untrained individual and a trained individual.



(3)

[illegible]

(c) Diabetes is a disease that affects millions of people across the world.

Human insulin is a small molecule that consists of two polypeptide chains linked by disulfide bridges.

Insulin from animals can be used to treat diabetes.

Human insulin can now be synthesised using genetically modified bacteria.

- (i) Describe how enzymes can be used to genetically modify the bacteria to produce human insulin.

(3)

(ii) Suggest **two** advantages of producing **human** insulin using genetically modified bacteria compared with insulin produced from animals.

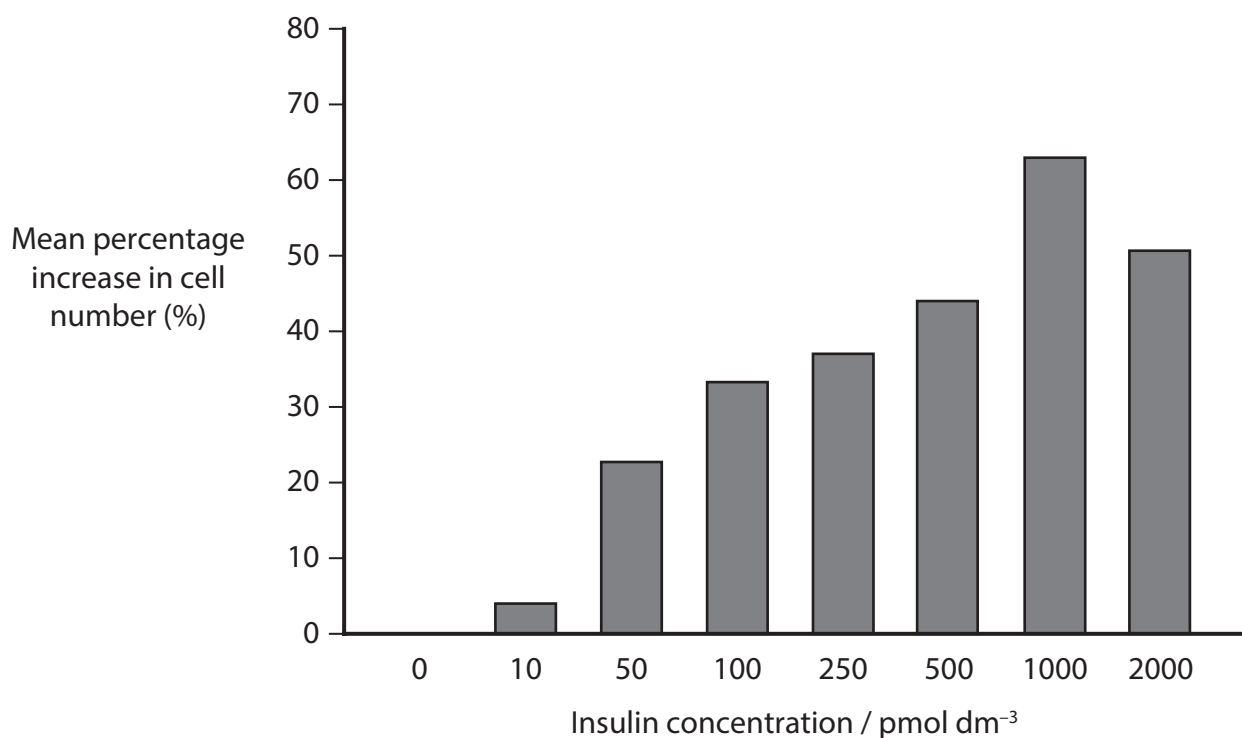
(2)

(d) Insulin is a peptide hormone that can stimulate cell division.

In an investigation, cells were incubated in solutions containing different concentrations of insulin.

After 48 hours, the increase in cell number was measured.

The results are shown in the graph.



(i) Give **one** reason why there is no bar on the graph for an insulin concentration at 0 pmol dm^{-3} .

(1)

(ii) Describe how the investigation was designed so that error bars could be produced.

(2)

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(iii) State how the error bars can be used to interpret the results of this investigation.

(1)

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(Total for Question 5 = 13 marks)